Practical No. 01

a) Develop an Android application implementing AppBar (Toolbar) and Options Menu.

Project Name: ToolbarMenuDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:paddingTop="25dp"
        android:theme="?attr/actionBarTheme"
        app:title="Toolbar Demo"
        app:titleTextColor="@android:color/white" />
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="AppBar and Options Menu Example"
        android:textSize="20dp"
        android:padding="22dp"/>
</LinearLayout>
```

Filename: menu_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">


    <item
        android:id="@+id/menu_home"
        android:title="Home"
        app:showAsAction="never"/>

    <item
        android:id="@+id/menu_about"
        android:title="About"
        app:showAsAction="never"/>
    <item
        android:id="@+id/menu_logout"
        android:title="Logout"
```

```
        app:showAsAction="never"/>
    <item
        android:id="@+id/menu_settings"
        android:title="settings"
        app:showAsAction="never"/>


</menu>
```

Filename: MainActivity.java

```
package com.example.toolbarmenudemo;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.widget.Toolbar;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    Toolbar toolbar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        toolbar=findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);

    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu){
        getMenuInflater().inflate(R.menu.menu_main,menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item){
        int id=item.getItemId();

        if(id==R.id.menu_home){
            Toast.makeText(this,"Home Selected",Toast.LENGTH_SHORT).show();
        } else if (id==R.id.menu_settings) {
            Toast.makeText(this,"Settings Selected",
                    Toast.LENGTH_SHORT).show();
```

```
        } else if (id==R.id.menu_about) {
            Toast.makeText(this,"About Selected",Toast.LENGTH_SHORT).show();


        } else if (id==R.id.menu_logout) {
            Toast.makeText(this, "Logout Selected", Toast.LENGTH_SHORT).show();


        }
        return true;
    }
}
```

b). Develop an Android application demonstrating AlertDialog and custom DialogFragment.

ProjectName: DialogDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <TextView
        android:id="@+id/txtContext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Long Press Me"
        android:textSize="24sp"
        android:padding="20dp"/>
    <Button
        android:id="@+id/btnPopup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Popup Menu"
        android:layout_marginTop="30dp"/>
</LinearLayout>
```

Filename: dialog_custom.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:background="#FFFFFF"
        android:text="Login karoo"
        android:textSize="22sp"
        android:textStyle="bold" />
    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter User Name"
        android:layout_marginTop="15dp"/>

    <EditText
        android:id="@+id/etName1"
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:layout_marginTop="15dp"/>
</LinearLayout>
```

Filename: CustomDialogFragment.java

```
package com.example.dialogdemo;

import android.app.AlertDialog;
import android.app.Dialog;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.fragment.app.DialogFragment;

public class CustomDialogFragment extends DialogFragment {

    EditText etName;

    @NonNull
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {

        AlertDialog.Builder builder =
                new AlertDialog.Builder(requireActivity());

        LayoutInflater inflater =
                requireActivity().getLayoutInflater();

        View view = inflater.inflate(R.layout.dialog_custom, null);

        etName = view.findViewById(R.id.etName);

        builder.setView(view)
                .setTitle("Input Dialog")
                .setPositiveButton("Submit", (dialog, which) -> {

                    String name = etName.getText().toString();

                    Toast.makeText(
                            requireActivity(),
                            "Entered: " + name,
                            Toast.LENGTH_SHORT
                    ).show();

                })
                .setNegativeButton("Cancel",
                        (dialog, which) -> dialog.dismiss());

        return builder.create();
```

```
    }
}
```

Filename: MainActivity.java

```
package com.example.dialogdemo;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnAlert, btnCustom;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        btnAlert = findViewById(R.id.btnAlert);
        btnCustom = findViewById(R.id.btnCustom);

        // AlertDialog
        btnAlert.setOnClickListener(v -> {

            AlertDialog.Builder builder =
                    new AlertDialog.Builder(this);

            builder.setTitle("Delete File")
                    .setMessage("Are you sure you want to delete?")

                    .setPositiveButton("YES", (dialog, which) -> {

                        Toast.makeText(
                                this,
                                "File Deleted",
                                Toast.LENGTH_SHORT
                        ).show();

                    })

                    .setNegativeButton("NO",
                            (dialog, which) -> dialog.dismiss());

            builder.show();
        });

        // Custom DialogFragment
        btnCustom.setOnClickListener(v -> {

            CustomDialogFragment dialog =
```

```
                    new CustomDialogFragment();

            dialog.show(
                    getSupportFragmentManager(),
                    "CustomDialog"
            );
        });
    }
}
```

c) Develop an Android application implementing contextual menus and popup menus.

ProjectName: MenuDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">
    <TextView
        android:id="@+id/txtContext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Long Press Me"
        android:textSize="24sp"
        android:padding="20dp"/>
    <Button
        android:id="@+id/btnPopup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Popup Menu"
        android:layout_marginTop="30dp"/>
</LinearLayout>
```

File Name: menu_options

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/menu_edit" android:title="Edit"/>
    <item
        android:id="@+id/menu_delete"
        android:title="Delete"/>
    <item
        android:id="@+id/menu_share"
        android:title="Share"/>
</menu>
```

File Name: MainActivity.java

```
package com.example.menudemo;
import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.PopupMenu;
```

```
import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


    TextView txtContext;
    Button btnPopup;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        txtContext = findViewById(R.id.txtContext);
        btnPopup = findViewById(R.id.btnPopup);


        // Register Context Menu
        registerForContextMenu(txtContext);


        // Popup Menu
        btnPopup.setOnClickListener(v -> {


            PopupMenu popupMenu = new PopupMenu(MainActivity.this, btnPopup);


            popupMenu.getMenuInflater().inflate(
                    R.menu.menu_options,
                    popupMenu.getMenu());


            popupMenu.setOnMenuItemClickListener(item -> {
                Toast.makeText(MainActivity.this,
                        item.getTitle(),
                        Toast.LENGTH_SHORT).show();
                return true;
            });


            popupMenu.show();
        });
    }


    // Context Menu Creation
    @Override
    public void onCreateContextMenu(ContextMenu menu,
                                    View v,
```

```
                                        ContextMenu.ContextMenuInfo menuInfo) {


        super.onCreateContextMenu(menu, v, menuInfo);


        menu.setHeaderTitle("Select Option");
        getMenuInflater().inflate(R.menu.menu_options, menu);
    }


    // Context Menu Click Handling
    @Override
    public boolean onContextItemSelected(@NonNull MenuItem item) {


        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();


        return true;
    }
}
```

d) Develop an Android application implementing Explicit and Implicit Intents.

ProjectName: IntentDemo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp"
    >

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Explicit Intent" />

    <Button
        android:id="@+id/btnImplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Implicit Intent" />

</LinearLayout>
```

Filename: activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/main"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second ACtivity"
        android:textSize="24sp"/>


</LinearLayout>
```

Filename:Main Activity.java

```
package com.example.intentdemo;

import android.content.Intent;
import android.os.Bundle;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit,btnImplicit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicit = findViewById(R.id.btnImplicit);

        btnExplicit.setOnClickListener(v->{
            Intent intent = new Intent(MainActivity.this,SecondActivity.class);
            startActivity(intent);
        });

        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(
                    Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com")
            );
            startActivity(intent);

        });
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;


        });
    }
}
```

Filename: SecondActivity.java

```
package com.example.intentdemo;

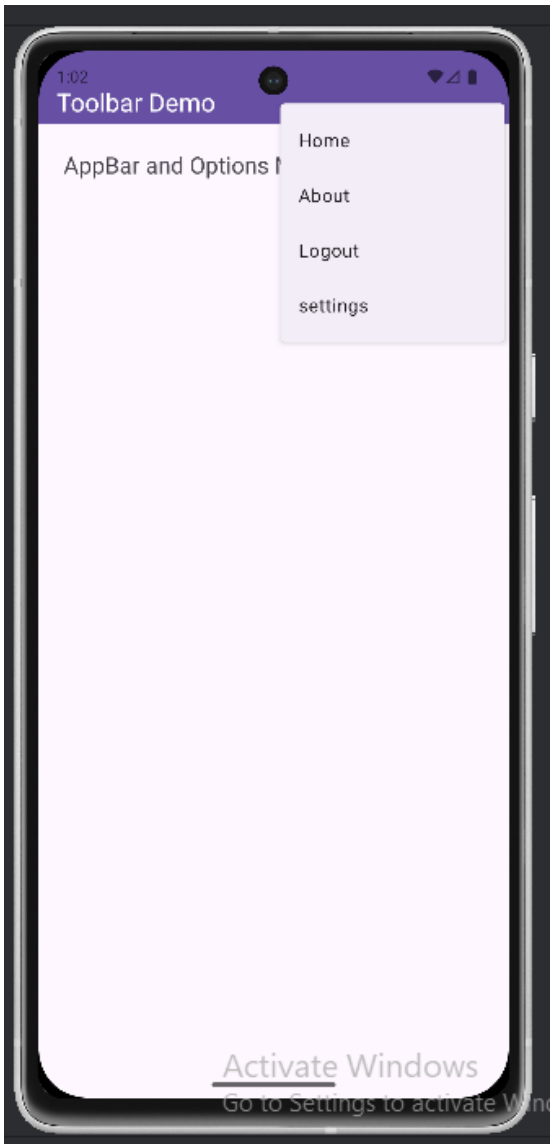
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

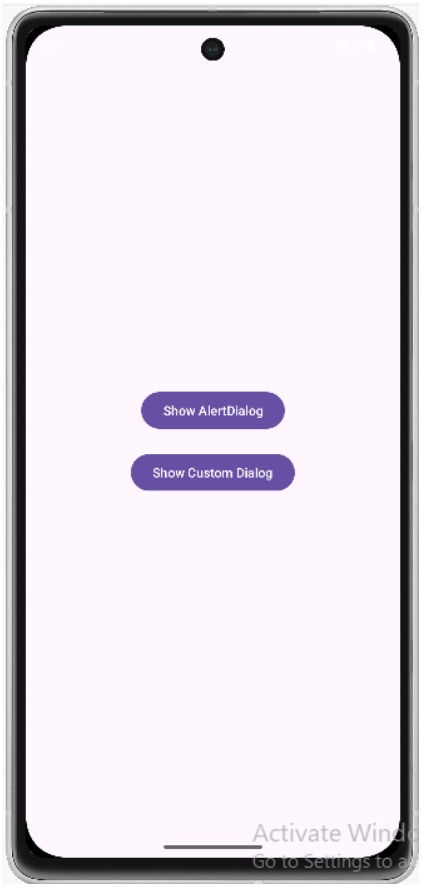
public class SecondActivity extends AppCompatActivity {

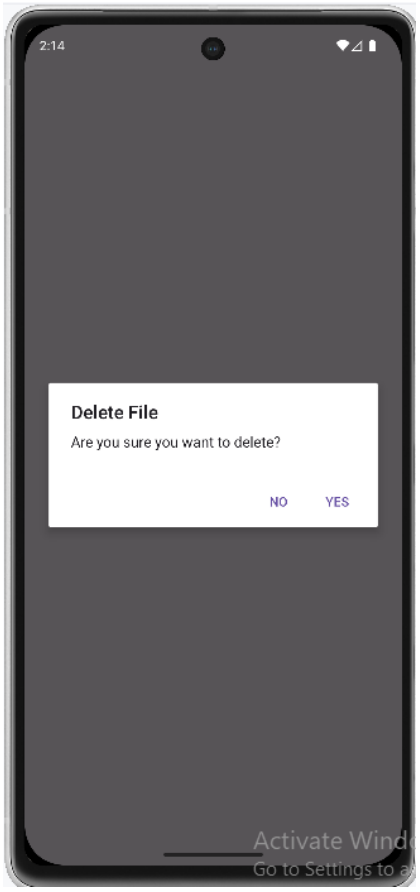
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_second);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });
    }
}
```

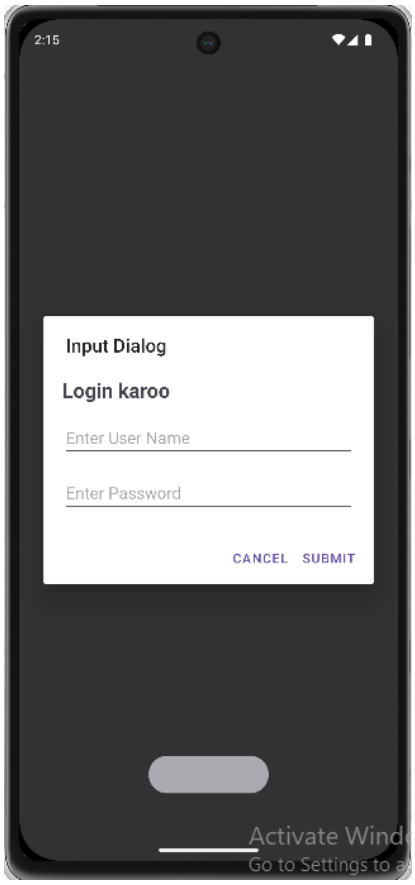
a) Output:


1:02
Toolbar Demo
AppBar and Options
Home
About
Logout
settings
Activate Windows

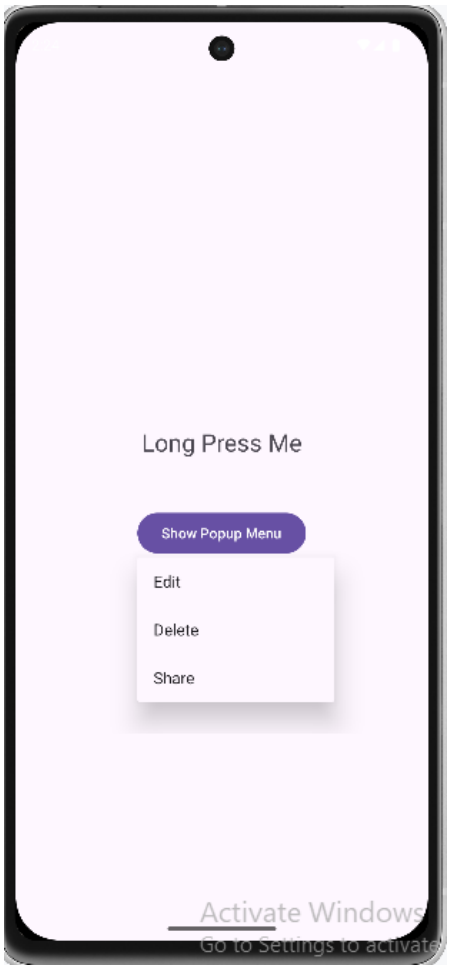
b) **Output:**


Show AlertDialog
Show Custom Dialog

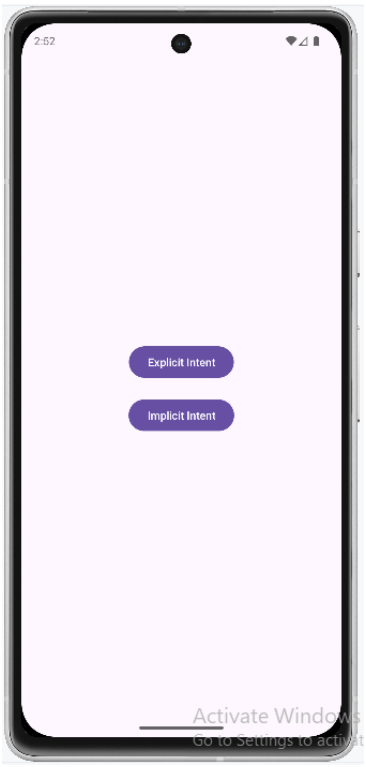

2:14
Delete File
Are you sure you want to delete?
NO
YES

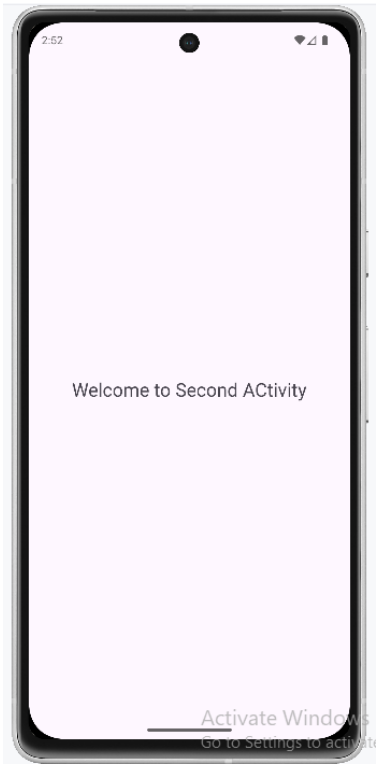

2:15
Input Dialog
Login karoo
Enter User Name
Enter Password
CANCEL
SUBMIT

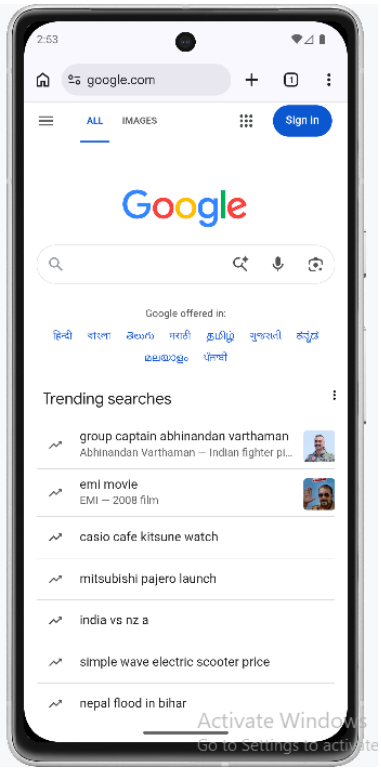
c) Output:


Long Press Me
Show Popup Menu
Edit
Delete
Share
Activate Windows
Go to Settings to activate

d) Output:


2:52
Explicit Intent
Implicit Intent


2:52
Welcome to Second ACtivity


2:53
google.com
ALL
IMAGES
Sign in
Google
Google offered in:
Trending searches
group captain abhinandan varthaman
Abhinandan Varthaman – Indian fighter pi...
emi movie
EMI – 2008 film
casio cafe kitsune watch
mitsubishi pajero launch
india vs nz a
simple wave electric scooter price
nepal flood in bihar